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Smart Attendance: Agentic AI-Powered Face Authentication with Anti-Spoofing

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Abstract—Existing classroom attendance systems rely on manual attendance or single-factor biometric matching, and are vulnerable to proxy attendance, spoofing and ineptitude. This paper proposes an intelligent attendance system that combines facial recognition and a multi-stage anti-spoofing process in a web environment, to offer a secure and fast attendance system. This framework is based on a three-tier architecture with a web-based user interface, Flask-based server and SQLite database for storing user data, time-table and attendance. The system employs client-side object detection to prevent mobile phone spoofing before capturing the image; server-side screen and geometric analysis to detect display spoofing. In addition, the facial liveness is confirmed by detecting blinks and limited head movements using facial landmarks. Matching to authenticate a user is done using cosine similarity of deep facial embeddings. An attendance controller based on time intervals also avoids duplicate attendance within a time interval. Experiments demonstrate that multi-stage verification improves its performance, prevents malicious attempts to record attendance and enables its use in non-critical academic settings. The framework supports secure student registrations, integrated with lecture schedule and monitoring through an administrator console that will allow deployment in institutional attendance management systems with improved transparency and accountability.

Index Terms—Smart Attendance System, Face Recognition, Anti-Spoofing, Liveness Detection, MediaPipe FaceMesh, VGG-Face, Deep Learning, Biometric Authentication, Period-Based Attendance Control

I. INTRODUCTION

Accurate attendance tracking plays a fundamental role in maintaining academic discipline, evaluating student participation, and supporting institutional administration. In many educational environments, attendance is still recorded manually or through semi-automated systems that depend on basic authentication techniques such as identity cards, passwords, or single-stage biometric verification. Although these approaches are simple to implement, they repeatedly suffer from serious limitations covering proxy attendance, identity misuse, and lack of real-time verification. As digital learning environments continue to expand, institutions require secure and intelligent attendance solutions that make sure only physically present students are recorded as participants in a classroom session.

Facial recognition technology is becoming a promising tool for attendance automation as it enables identity verification without contact and with off-the-shelf camera devices. Use of hardware infrastructure: Compared to fingerprint or smart card readers, facial authentication is more user-friendly and less reliant on other hardware infrastructure. However, traditional face recognition attendance systems are still challenged with spoofing attacks in which attackers try to cheat authentication with a printed photo, a recorded video, or an image on an electronic display. These weaknesses undermine confidence in automated attendance systems, especially in cases where attendance is critical, and suggest the development of novel verification methods that integrate one's identity with a proof of physical presence.

Recent advances in AI and CV further allow combing many verification approaches along a single pipeline, yielding better detection performance than any single-model solutions. Specifically, the combination of object detection, facial landmark tracking, and embedding-based identity matching will enable systems to perform dual verification, i.e., who the user is and if the user is actually in front of the camera. Such a multi-layered verification methodology also enhances robustness for real-world academic applications where recognition performance may be subject to illumination changes, device variety, and varying user habits.

Towards that end, this work presents intelligent attendance system that combines agentic artificial intelligence with multi-layer anti-spoofing verification to offer a secure yet feasible attendance management system. The application is a 3-tier application: it contains a web-based interface at the top layer, a server-side processing tier using Flask framework, and a small database tier

where user's credentials and timetable and attendance data are stored. The browser layer collects webcam frames, conducts early-stage spoof detection, and then sends data to the server, so as to avoid wasting computation and improve overall efficiency. It is in the server tier where the main verification pipeline, including the device detection and the facial landmark tracking as well as the identity verification through deep feature comparison is executed. Then the attendance records are safe which can be used for any future doubts by being automatically captured in a timetable-aware database that is intelligent enough not to take multiple records in a specific academic duration.

One of the contributions of such a system is that it brings a sequential multi-stage anti-spoofing detection strategy for addressing a number of prevalent attack situations rooted in current attendance systems. The solution doesn't trust the facial recognition alone, instead it makes sure that a mobile phone or an LCD screen isn't being used to show a fake face to the camera. next performs facial liveness validation via eye blink and controlled head motion determination relying on facial landmark analysis. Then the system begins to authenticate the identity via deep facial embeddings. This combined approach enhances the reliability of authentication while providing a seamless user experience for classroom-style deployment.

Another significant aspect of the proposed framework is the timetable-aware attendance management policy. Traditional attendance systems typically include clock-hour-based limitations, allowing for two or more entries during a single class block if sessions cross hourly boundaries. To address the problem, the attendance system presented here is linked with predefined timetable slots rather than the rigid clock intervals. This is done with the stipulation that students can only take attendance once per each regularly scheduled academic session to increase record accuracy and to prevent accidental repetition.

In addition to enhancing the security of authentication, the system offers different interfaces for the administrator and other users (students), where they can register, configure the timetable, monitor the attendance, and view the history. Such capabilities allow the institution to handle attendance management procedures effectively, with full visibility and accountability over academic processes. By integrating accessibility features with multi-factor authentication and structured scheduling constraints, the proposed approach illustrates how emerging computer vision can be harnessed to improve the dependability of attendance management in educational settings.

The collective face recognition, liveness detection and device-level spoof prevention in a single architecture further demonstrates the promise of agentic AI pipelines in strengthening biometric attendance systems. Thus, the proposed scheme is a feasible solution for secure, automated, and scalable attendance monitoring in modern day academia.

II. RELATED WORK

Automated attendance systems have attracted much interest recently following the inadequacy of conventional manual attendance methods. Previously, a paper-based attendance system or scanning an ID card was considered as the attendance recording technique, which was slow and error-prone. In response, researchers have investigated biometric systems, including facial recognition, mobile authentication, and verification based on wireless devices, to enhance the accuracy and speed of attendance.

One line of work in automated attendance application is leveraging facial recognition technology for identity authentication. A robust real-time attendance monitoring system that utilizes facial recognition algorithms namely Histogram of Oriented Gradients (HOG) along with Haar Cascade classifiers and Convolutional Neural Networks (CNNs) has shown that automation can drastically reduce the manual work and also increase reliability in class room setup. Experimental assessment in this paper achieved a recognition accuracy of nearly 94.6 % by using advanced detection schemes, thus permitting envisioning the adoption of CV techniques for institutionalized attendance procedure (Patil & Shinde, 2020).

Likewise, studies on Android based attendance systems emphasize the necessity of implementing face detection and classification algorithms in mobile devices for eliminating the traditional means of manually recording. They generally operate on a pipeline including the stages of face detection, feature extraction, classifier training, and recognition. Algorithms like the Adaboost cascade classifiers and the Local Binary Pattern (LBP) feature descriptor are extensively utilized to detect facial areas and to do classification with good efficacy. This enabled mobile-friendly solutions to enhance flexibility and flexibility, whereas recording attendance through mobile devices ensures convenience for both the academic and the office environments [11 (Bai et al., 2020)].

An additional significant innovation in the field of attendance systems is the anti-spoofing features incorporated with facial recognition system. Traditional face recognition systems are still vulnerable to attacks with printed photo or digital display screen. To overcome this limitation, a Silent Face Anti-Spoofing framework was introduced based on the Convolutional Neural Networks along with Fourier Transform-based frequency analysis, to distinguish between real and plain faces. The method was trained on a large-scale dataset of around 15 million images and experiment results showed better detection performance for spoofing attacks with different illumination and distance. The process of authenticating was significantly improved by distinguishing between a real human being and computer rendered images, especially in case of attendance where he states: "The study's rms authentication reliability by differentiating live face from photoshopped images in attendance system was substantially increased" (Salam et al., 2025).

Besides identity verification, recent studies have investigated attention-aware attendance systems which capture attendance and student engagement simultaneously. For example, a classroom attendance system used face recognition matched with gaze tracking and behavioral analysis to assess whether students were really paying attention in class. Attendance was registered only if the attention value was higher than a certain threshold, so as to make the tracking of class participation more efficient. Merging OpenCV-based real time processing with KNN classification allowed to efficiently recognize the subject in real time in challenging conditions of classroom, confirming the feasibility of biometric recognition under behavior analysis (Thakur et al., 2025).

In addition to vision-based attendance methods, a number of scholars have studied wireless communication technology based location-aware attendance authentication. Students can identify themselves with their mobile devices only once per class when they are physically in the classroom in BLE beacon–based attendance models. For attendance submission in such systems, beacon signals radiate unique identifiers, so that the attendance submission application knows which one it is receiving attendance from, making fraudulent remote submissions impractical. However, these methods still rely on external devices and do not perform identity confirmation with biometric features (Noguchi et al., 2015).

Despite these advancements, a majority of the present attendance systems still make use of single-layer authentication mechanism (e.g., face detection or device proximity verification) and do not incorporate full-fledged anti-spoofing techniques. Although Convolutional Neural Network-based (CNN—based) facial recognition offers higher accuracy when compared to conventional biometric schemes, it is still susceptible to impersonation attacks that exploit printed photos, LCD displays or videos. In addition, many systems focus on engagement monitoring or location verification alone without integrating a set of verifications into a single authentication pipeline.

To overcome these limitations, the system proposes a multi-layer agentic verification architecture that integrates object detection based spoof prevention, facial landmark based liveness verification and deep embedding based identity matching under one attendance framework. The system leverages sequential authentication protocols with timetable-aware attendance management to ensure robustness, while suited for academic settings. This multilayer methodology separates the presented model from previous work on attendance automation where only one biometric or proximity based verification were exploited, and shows that multiple computer vision solutions can be combined to securely authenticate attendance.

III. PROPOSED FRAMEWORK

The proposed framework presents an intelligent attendance management system based on face recognition with a multi-layer anti-spoofing pipeline for secure and robust student authentication. In contrast to traditional approaches that depend solely on facial recognition or human verification, the system described in this article is based on agentic artificial intelligence with several verification stages. They are united in verifying device authenticity, facial liveness and identity matching prior to registering attendance.

The overall framework consists of three major components:

1. Client-side spoof detection and frame acquisition
2. Server-side liveness verification and identity authentication
3. Period-aware attendance recording and database management

The objective of this framework is to prevent fake attendance attempts while making sure that attendance is recorded only when the student is physically present and successfully verified through multiple authentication stages.

A. Overall Architecture

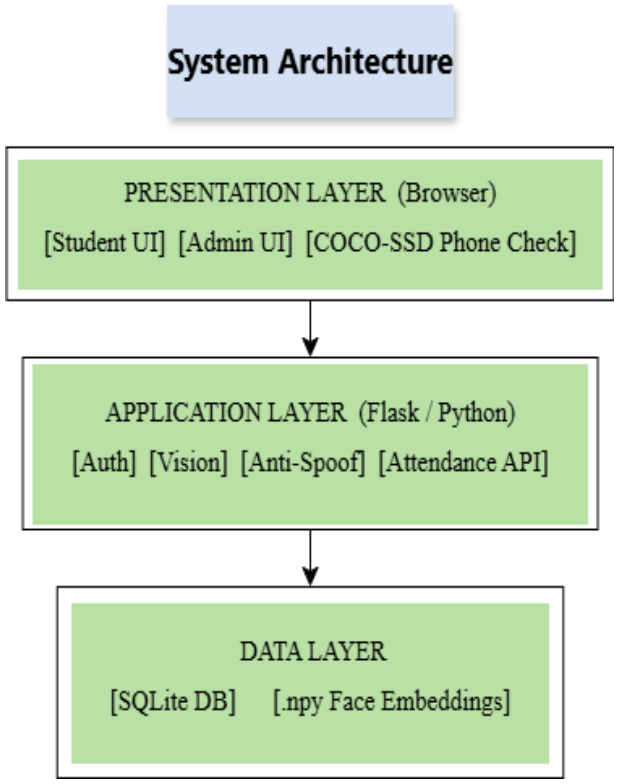


Fig. 1. Overall Architecture of the proposed framework.

The overall architecture follows a three-tier browser, server, data storage model. The browser layer collects webcam frames and does early spoof detection with lightweight object detection models. This avoids sending suspicious frames to the server unnecessarily and enhances the computational efficiency.

The captured frames are then sent to the application server, in which more verification steps are performed in a pipeline manner. These stages are display-based spoofing detection, facial landmark-based liveness verification, and embedding-based identity recognition. Once all the verification process are successfully passed through, only then it stores the attendance in the database system.

The final tier in the architecture stores and manages the attendance records, student credentials, and time table details in a structured relational database. Such a multi-tiered architecture guarantees scalability, security and effective running of the attendance verification pipeline.

B. Flow Diagram

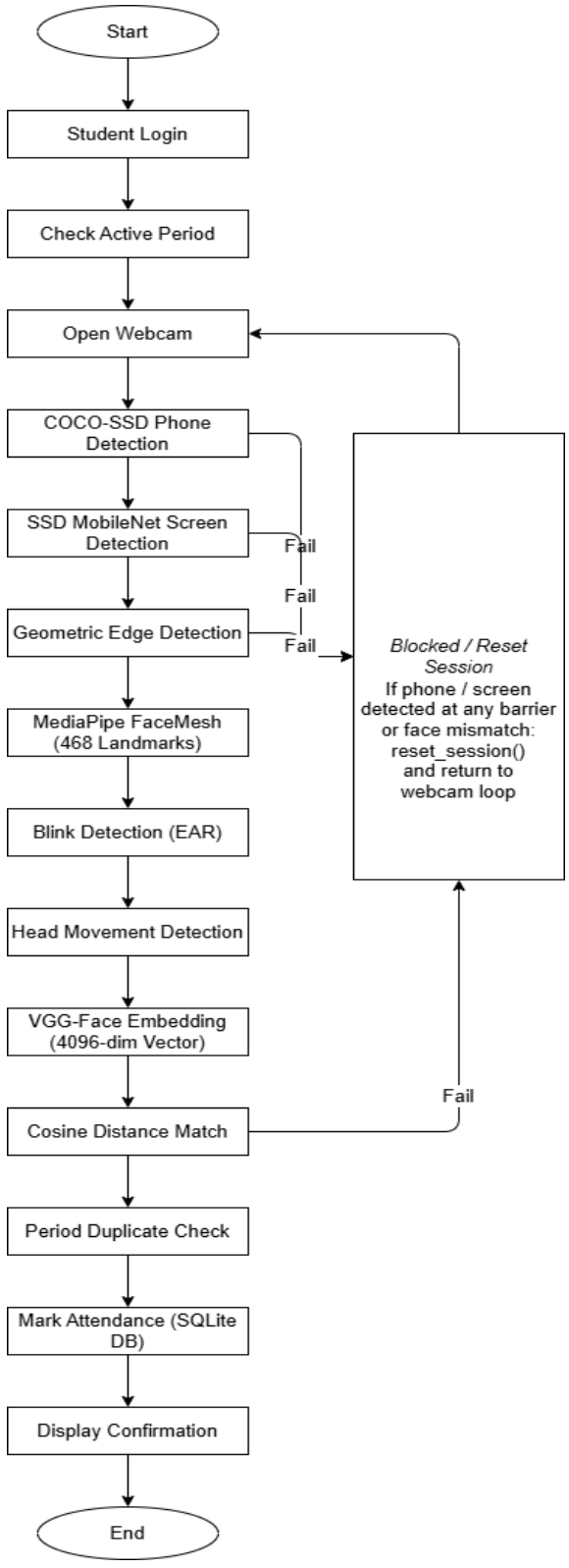


Fig. 2. Flow diagram of the proposed framework.

The operational workflow of the system follows a structured sequence of verification stages designed to prevent spoofing attacks at different layers. The procedure begins when a student logs into the attendance portal and access the webcam interface for identity verification. A frame capture loop is running to fetch images from the camera and do client-side object detection to find suspected spoof objects (mobile devices).

If no spoofing device is detected, the captured frame is sent to the server for further processing. The server performs screen detection based on deep learning object detection models to screen out laptop displays or tablet screens used for spoofing attacks. Subsequently, classical geometric contour detection methods are employed to find partial screen borders that possibly cannot be detected by neural network-based detection methods.

After successful spoof filtering, facial landmark detection is run to perform liveness verification (using blink detection, head moving tracking). These activities are proof the face that you are detected is from a real person, not a still photo. Lastly, the facial embeddings extracted from the live frame is matched with the saved embeddings of the logged-in student to verify the identity prior to marking the attendance.

C. *Client-Side Device Detection Module*

First, the proposed framework runs in the browser and executes a lightweight object detector to perform live spoofing detection. This mod-ule detects whether a mobile phone is displaying a face in front of the webcam. Block the frame transmission immediately if an anomaly is found.

Two main benefits are derived from running the spoof detection on the client side at the PC. Firstly, it prevents the waste of the server's computational resources by preliminarily discarding the invalid frames in the pipeline. And, it stops spoofing attempts at an earlier stage before consuming any resources in the authentication process, which contributes to system efficiency and user experience.

D. *Server-Side Screen Detection Module*

The second part of the framework applies screen detection at the server side by leveraging deep learning-based object recognition with geometric contour analysis. This component also detects laptop screens, tablets, or satins that may be used to circumvent authentication methods.

The deep learning detector detects monitor-like regions in the obtained frames, and the contour-based geometric checking verifies if a rectangular screen area can be found even when some borders are missing. This twofold detection approach increases the robustness to display-based spoofing attacks, which are often overlooked by single-model solutions.

E. *Facial Liveness Verification Module*

Following the completion of the spoof detection stages, the system carries out facial liveness verification through the application of facial landmark analysis. It analyzes eye movement and head rotation sequences to verify that the identified face is that of a real person.

Blink detection is implemented by eye landmark based measurements that qualify the eye opening changes in consecutive frames. In the same way, the head movement validation is done by comparing the nose-tip coordinates in the next frames. Attendance verification proceeds only when both blinking and directional head movements are successfully detected, ensuring reliable confirmation of physical presence.

F. *Face Recognition and Attendance Recording Module*

Lastly, the fourth step of the system is the identity-verification via deep facial embedding comparison. The feature vector that is extracted from the live webcam frame is matched with the saved embedding of the authenticated student using a similarity measure. Attendance is only taken when the similarity score exceeds the preset matching threshold.

Once identity verification is successfully done, the timetable-mindful recording technique will save the attendance record into the database. Attendance can only be marked once in each academic period under this procedure and would not be able to submit multiple attendance records for the same session. The stored information consists of student identity, subject, time-stamp values, and verification status.

By combining sequential spoofing detection, liveness verification and embedding-based recognition, the proposed framework demonstrates a reliable and effective solution for auto-mated attendance system management in futuristic educational environment.

IV. RESULTS

A. *Spoof Detection Performance*

The anti-spoofing module is evaluated across two test sets comprising 60 trials in total. The first set consisting of 30 device-based spoofing attempts using

physical phones and laptop screens displaying registered face images. The second set consisting 30 printed photograph attacks using A4-sized printouts. Results are summarized in Table 1.

Test Set	Attack Type	Trials	TP	FN	Detection Rate
Set A	Device (Phone & Laptop)	30	21	9	70.0%
Set B	Printed Photograph	30	26	4	86.7%
Overall	**Combined**	**60**	**47**	**13**	**78.3%**

Table. 1. Spoof detection performance.

The system exhibited more accurate detection results for printed photograph attacks, achieving a detection rate of 86.7% versus 70.0% for device-based attacks. The slightly lower detection rate for device-based spoofing is due to different screen brightness levels and angles of phone and laptop were used to be displayed. The geometric contour based detector and the SSD MobileNet combined to yield a true acceptance rate of 78.3% on the overall spoofing detection across 60 trials, with a miss rate of 21.7%.

B. Identity Recognition Accuracy

The face recognition module was evaluated using 44 trials — 22 genuine attempts by registered students and 22 impostor attempts by unregistered individuals. VGG-Face embeddings were matched using cosine distance with a threshold of 0.72. Results are presented in the Table-2.

	Predicted: Accept	**Predicted: Reject**
Actual: Genuine	TA = 18	FR = 4
Actual: Impostor	FA = 0	TR = 22

Metric	Formula	Value
Genuine Acceptance Rate (GAR)	TA / (TA+FR) × 100	**81.8%**
False Rejection Rate (FRR)	FR / (TA+FR) × 100	**18.2%**
False Acceptance Rate (FAR)	FA / (FA+TR) × 100	**0.0%**
True Rejection Rate (TRR)	TR / (FA+TR) × 100	**100.0%**
Overall Accuracy	(TA+TR) / Total × 100	**90.9%**

Table. 2. Identity recognition accuracy.

The VGG-Face recognition module achieved a False Acceptance Rate (FAR) of 0%, meaning no impostor was ever granted attendance — a critical security requirement for the system. The Genuine Acceptance Rate (GAR) was 81.8%, with 4 genuine students being incorrectly rejected, likely due to variations in lighting and facial angle at the time of attendance compared to registration. The overall system accuracy across all 44 trials was 90.9%.

V. RESULTS AND DISCUSSION

The experimental Evaluation of the proposed agentic attendance authentication framework in our-bench for both static and dynamic faces shows that integrating multi-layer spoof detection with facial landmark–based liveness verification offers a higher security level compared to traditional solutions based only on static face recognition while maintaining a comparable processing rate. The client-side object detection module successfully prevents most of the mobile phone–based spoofing attacks, as frames are not sent to the server, which in turn decreases the server's computation and makes the system more responsive. Moreover, the server-side module detects additional spoofing attacks with the use of laptop screens and tablets, while geometric contour detection provides an additional layer of protection against partially visible display-based spoofing attacks.

The facial landmark–based liveness verification ensures that authentication is only per-formed when natural eye blinking and bidirectional head movement (left/right) are observed, which acts as an indicator of physical presence. Embedding-based identity verification allows for precise one-to-one comparison between live webcam frames and previously stored enrollment embeddings, effectively reducing complexity of recognition while preserving its robustness. In addition, the timetable-aware attendance validation algorithm enhances the accuracy of recording by prohibiting multiple attendance records for the same session. To conclude, the processing of the integration of spoof

detection, liveness verification, and embedding-based recognition enable an effective and se- cure solution for automated attendance management in academic applications.

VI. FUTURE WORK

Although the proposed attendance authentication scheme shows consistent performance with normal classroom environment, several enhancements and extensions are feasible for the system.

- **Integration with Multi-User Session Handling:** Attendance processing and verification in the current implementation retains global liveness verification states, which prohibits universal multi-user authentication. Future improvements might include session-specific state to enable concurrent attendance verification for many users within a large-classroom setting.
- **Deployment with Cloud-Based Storage Systems:** The current system locally stores the facial embeddings through structured file storage along with a lightweight relational database. It is anticipated that future releases of the framework will incorporate cloud storage solutions to enhance scalability and allow for centralized attendance management for a number of organizations or branches.
- **GPU-Based Real-Time Optimization:** Although the system runs efficiently on commodity CPU-based hardware, adding GPU acceleration can provide additional speedup for the deep embedding extraction and the object detection modules. This enhancement would enable support for large scale class room deployments, with multiple concurrent verification requests.
- **Integration with Email or OTP-Based Recovery Systems:** The current password recovery mechanism relies on student ID numbers for authentication. Future implementations can incorporate secure email-based or one-time password (OTP) recovery systems to strengthen account security and improve usability.

Overall, these improvements aim to enhance system scalability, usability, and authentication robustness while supporting broader deployment of intelligent attendance monitoring systems in educational environments.

VII. CONCLUSION

This paper presented an agentic artificial intelligence–based attendance authentication framework designed for improved reliability and security of automated attendance systems through multi-layer verification mechanisms. Our system combines client-side device detection, server-side spoof filtering, face landmark-based liveness check, and deep embedding-based identity recognition in a single pipeline that verifies identity and confirms physical presence prior to logging attendance.

Moreover, the framework adds a timetable-aware attendance validation mechanism to avoid recording two attendance entries within the same academic session, the weakest point of the conventional attendance system based on clock hours. According to the experimental results, the combined scheme of blink detection, head motion tracking and embedding-based similarity comparison enhances the security as well as the performance without the need of any special hardware setup.

Overall, by integrating multi-layered spoof detection, facial landmark and deep embedding-based recognition, our solution achieves practical and scalable secure attendances. The developed framework can catalyze next-generation intelligent attendance systems based on computer vision to improve reliability, transparency and utilization in academic attendance management solutions.

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